Mathematical Summary of pyMacroIO: Dynamic Disequilibrium Model with Input-Output Structure

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This document gives an overview of pyMacroIO, a Dynamic Disequilibrium Model with Input-Output Structure. The model uses an input-output table as its structural backbone but is best understood as a dynamic disequilibrium macroeconomic model: production is supply- and capacity-constrained rather than demand-determined, and quantity rationing takes the place of price-mediated market clearing. It draws on the formulation in [1,2,3] but extends these implementations substantially. The model is discrete-time and sector-level. It admits five production functions (traditional Leontief, adapted Leontief, linear, CES, KLEMS), tracks per-sector inventories, includes labour hiring and firing, a Muellbauer consumption rule, exogenous consumption and supply-side shocks, and piecewise-constant technical change. Further components are an optional endogenous price pass-through with asymmetric cost transmission, a multi-region structure with R regions, three trade closures (import flexibility, sourced supplement, export pull), two endogenous final demand rules (income-indexed government spending and a Keynesian savings-to-investment closure), an exogenous time-varying labour supply path, a Blanchflower-Oswald wage curve with optional downward nominal wage rigidity [6], and skill-stratified household savings rates. Sections 1 to 8 cover the single-region formulation. Section 9 covers technical change, Section 10 the multi-region structure, Section 11 the trade closures, and Section 12 the optional price pass-through. Base-year data are taken from EXIOBASE 3 [4].

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1 Model Structure

1.1 Basic Setup

The formulation is defined over N sectors and T time periods. In the single-region case N is the number of sectors in one region. In the multi-region case $N = \sum_r N_r$ spans all regions (see Section 10). Gross output at time t is denoted by $\mathbf{x}_t \in \mathbb{R}^N$, the technical coefficient matrix by $\mathbf{A} \in \mathbb{R}^{N \times N}$, total demand by $\mathbf{d}_t \in \mathbb{R}^N$, and final demand by $\mathbf{f}_t \in \mathbb{R}^N$. An optional endogenous price index $\mathbf{p}_t \in \mathbb{R}^N$, normalised to $\mathbf{p}_0 = \mathbf{1}$ in the base year, is defined in Section 12.

1.2 Economic Balance Constraint

The fundamental static balance is

$$\mathbf{x} = \mathbf{A}\mathbf{x} + \mathbf{f}, \tag{1}$$

from which

$$\mathbf{x} = (\mathbf{I} - \mathbf{A})^{-1}\mathbf{f} = \mathbf{L}\mathbf{f} \tag{2}$$

is obtained, where $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$ is the Leontief inverse. In the dynamic model this structure underlies calibration and the identification of essential inputs. Within-period equilibrium is enforced by the sequential determination of labour, demand, production, intermediate consumption, and inventory update.

2 Data and Parameter Calibration

2.1 Technical Coefficient Matrix

The technical coefficient matrix $\mathbf{A}$ is constructed from base-year intermediate flows $\mathbf{Z}_0$ and base-year gross output $\mathbf{x}_0$. The gross output $\mathbf{x}_0$ is obtained from the row identity: the output of each sector equals the sum of its intermediate sales plus total final demand. Then

$$A_{ij} = \frac{Z_{0,ij}}{x_{j,0}}, \tag{3}$$

with the convention that each column sum of intermediate inputs is less than output:

$$\sum_{i=1}^{N} A_{ij} < 1 \quad \forall j. \tag{4}$$

2.2 Essential Inputs Identification

For the adapted Leontief production function, a binary matrix $\mathbf{A}_{\text{essential}}$ classifies each input as essential or non-essential. An input i to sector j is essential if supplier i has high forward linkage and purchaser j has high backward linkage. Let $\mathbf{B} = \hat{\mathbf{x}}_0^{-1}\mathbf{Z}_0$ be the allocation coefficient matrix, in which each flow is normalised by the supplier's gross output ($B_{ij} = Z_{0,ij}/x_{i,0}$), and let $\mathbf{G} = (\mathbf{I}-\mathbf{B})^{-1}$ be the Ghosh inverse. The forward linkage of supplier i is the ith row sum of $\mathbf{G}$, and the backward linkage of purchaser j is the jth column sum of the Leontief inverse $\mathbf{L} = (\mathbf{I} - \mathbf{A})^{-1}$, the Rasmussen measure. The combined linkage score is

$$\Lambda_{ij} = \frac{\sum_k G_{ik}}{\sum_k G_{ik}} \times \frac{\sum_k L_{kj}}{\sum_k L_{kj}}, \tag{5}$$

and the binary indicator is

$$A_{\text{essential},ij} = \mathbb{1}\big[\Lambda_{ij} > \theta\big] \cdot \mathbb{1}\big[A_{ij} > \theta_{\text{coef}}\big], \tag{6}$$

where $\theta > 0$ is a linkage threshold [5] and $\theta_{\text{coef}} > 0$ is a coefficient threshold that excludes trace technical coefficients from the essential set. The overbars denote means over the active set $\mathcal{A} = \{i : x_{i,0} > 0\}$. Region-sector cells with no activity are not sectors of the economy, so they enter neither mean and are excluded from the essential set, leaving both indices those of the active economy.

2.3 Value Added Decomposition

Value added in the base year is decomposed as

$$\mathrm{VA}_j = L_j + \mathrm{cap}_j + \mathrm{tax}_j + \mathrm{other}_j + \Pi_j, \tag{7}$$

where L_j is labour cost, cap_j capital cost, tax_j taxes, other_j other costs, and Π_j profits. Value added is taken from the column identity (output minus domestic intermediate inputs minus imports). Base-year labour $\ell_{j,0}$, capital shares λ_j^{cap}, tax shares λ_j^{tax}, and import shares λ_j^{imp} are used, so that profits are computed as

$$\Pi_{j,t} = x_{j,t} - \sum_{i=1}^{N} Z_{ij,t} - \ell_{j,t} - (\lambda_j^{\text{cap}} + \lambda_j^{\text{tax}} + \lambda_j^{\text{imp}})\, x_{j,t}. \tag{8}$$

3 Dynamic Model Equations

3.1 Labour Adjustment Mechanism

Labour is adjusted towards capacity and demand or input constraints via sector-specific hiring and firing speeds $\gamma_{\text{hire},j}$ and $\gamma_{\text{fire},j}$. When hiring and firing are switched off, $\ell_{j,t} = \ell_{j,t-1}$ is imposed. An exogenous labour supply path $\ell_{j,t}^{\text{ref}}$ sets the capacity reference for each period; when no schedule is prescribed, $\ell_{j,t}^{\text{ref}} = \ell_{j,0}$ for all t. Let $\delta_{j,t} \in [0, 1)$ denote the capacity reduction imposed by an active output-constraint shock on sector j (zero otherwise, as defined in Section 5). The following quantities are used.

$$\ell_{\text{cap},j,t} = \ell_{j,t}^{\text{ref}} \cdot \max\big(\min(1 - \delta_{j,t}, 1), \underline{\delta}\big), \tag{9}$$

$$\ell_{\text{cap,max},j,t} = \overline{\varsigma}\, \ell_{j,t}^{\text{ref}}, \quad \ell_{\text{cap,min},j} = \underline{\varsigma}\, \ell_{j,0}, \tag{10}$$

$$\ell_{\text{cap},j,t}^{\text{new}} = \begin{cases} \min\big(\ell_{\text{cap,max},j,t},\ \max(\ell_{\text{cap},j,t}, \ell_{\text{cap,min},j}, \ell_{j,t-1})\big) & \text{during a disruption shock to sector } j, \\ \ell_{\text{cap,max},j,t} & \text{otherwise,} \end{cases} \tag{11}$$

$$\sigma_{j,t} = \frac{\ell_{j,0}}{\phi_{j,t}^{L}\, x_{j,0}}, \tag{12}$$

$$x_j^{\text{pot}} = \min\big(x_j^{\text{inp}},\ d_j,\ \bar{x}_j^{\text{out}}\big), \quad \Delta_j = x_j^{\text{pot}} - \ell_{j,t-1}/\sigma_{j,t}, \tag{13}$$

where $\sigma_{j,t}$ is the labour-output ratio deflated by the cumulative labour-productivity factor $\phi^L_{j,t}$ (Section 5), with $\sigma_{j,t} = \ell_{j,0}/x_{j,0}$ when $\phi^L_{j,t} = 1$; x^{inp}_j is the input-constrained output potential (the binding inventory ratio across inputs); $\bar{x}^{\text{out}}_j$ is an exogenous output ceiling (default $+\infty$, set to a finite value to impose sector-level phaseout paths such as fossil fuel constraints from an energy model); and Δ_j is the gap between potential output and current labour capacity. The upper bound $\ell_{\text{cap,max},j,t}$ is time-varying because it tracks the exogenous supply path, whilst the floor $\ell_{\text{cap,min},j}$ is anchored to the base year. The labour adjustment uses $\gamma_{\text{hire},j}$ when $\Delta_j > 0$ and $\phi\,\gamma_{\text{fire},j}$ when $\Delta_j \le 0$,

$$\ell_{j,t} = \max\Big(\ell_{\text{cap,min},j},\ \min\big(\ell_{j,t-1} + \gamma_j\,\sigma_{j,t}\,\Delta_j,\ \ell^{\text{new}}_{\text{cap},j,t}\big)\Big), \tag{14}$$

where $\gamma_j = \gamma_{\text{hire},j}$ if $\Delta_j > 0$ and $\gamma_j = \phi\,\gamma_{\text{fire},j}$ otherwise, and $\phi \in (0, 1]$ is a firing-speed damping factor.

When the wage curve is active, a Blanchflower-Oswald adjustment modifies $\sigma_{j,t}$ before the desired-labour computation [6]. Let $U_{j,t-1}$ denote the sector-level unemployment rate in the previous period, supplied as an exogenous time series, and $\bar{U}_j$ the base-year value. The adjustment factor is

$$w_{\text{adj},j,t} = \left(\frac{\max(U_{j,t-1},\,\underline{u})}{\max(\bar{U}_j,\,\underline{u})}\right)^{-\beta^{\text{wc}}_j}, \tag{15}$$

where $\beta^{\text{wc}}_j > 0$ is the wage-curve elasticity and $\underline{u} > 0$ is a small positive floor that ensures the ratio is well-defined when unemployment approaches zero or is negative (the latter arising when labour demand exceeds supply). The effective ratio $\sigma^w_{j,t} = \sigma_{j,t} \cdot w_{\text{adj},j,t}$ replaces $\sigma_{j,t}$ in equation (14). When $U_{j,t-1} > \bar{U}_j$ (unemployment above the base-year level), $w_{\text{adj},j,t} < 1$ and wages are lower relative to base year; when $U_{j,t-1} < \bar{U}_j$, $w_{\text{adj},j,t} > 1$ and wages are higher.

When downward nominal wage rigidity is additionally active, the adjustment is bounded below,

$$w_{\text{adj},j,t} \ge \bar{\omega}, \tag{16}$$

where $\bar{\omega} \in (0, 1]$ is the wage-floor ratio, preventing wages from falling below a fraction $\bar{\omega}$ of their base-year level regardless of the unemployment rate. By default both features are inactive and the standard ratio $\sigma_{j,t}$ applies throughout.

3.2 Consumption Function (Muellbauer)

Aggregate consumption demand C_t is determined by a log-linear persistence rule expressed in terms of a *household consumption capacity* κ_t. The capacity is derived from disposable household income $\mathcal{Y}_t$ via

$$\kappa_t = \max\big((1 - s)\,(1 - \nu)\,\mathcal{Y}_t,\ \varepsilon_\kappa\big), \tag{17}$$

where $s \in [0, 1)$ is the household savings rate (inferred from base-year data if not set explicitly), ν is the extra-expenditure coefficient, and $\varepsilon_\kappa > 0$ is a small numerical floor. Disposable income is

$$\mathcal{Y}_t = \max\big(\beta \textstyle\sum_j \ell_{j,0} + (1 - \beta) \sum_j \ell_{j,t} + \sum_j \Pi_{j,t},\ \varepsilon_{\mathcal{Y}}\big), \tag{18}$$

where $\beta \in [0, 1]$ is the benefits parameter and $\sum_j \ell_{j,0}$ is base-year labour income. The benefits parameter stabilises labour income towards its base level, whilst profit income enters in full and is not smoothed. The base-year household income $\mathcal{Y}_0 = \max\big(\sum_j \ell_{j,0} + \sum_j \Pi_{j,0},\ \sum_j c_{j,0}/(1 - \nu)\big)$, the

production-side income floored at the level needed to fund base-year consumption, serves as the reference level in the closure modes below.

Three *household closure modes* are defined:

Return-to-base The log of the previous consumption ratio is set to zero (the ρ_1 term drops), so the aggregate converges back to the baseline whenever income recovers. The income argument to κ_t is replaced by the fixed signal $\mathcal{Y}_0\,\xi_t$ (base-year income scaled by the expectations trend); current-period wages and profits do not feed into consumption under this closure. Consequently, supply-side shocks propagate through inventory and capacity channels only, that is, the Keynesian income-demand multiplier is inactive. See Section 10.3 for the full multi-region specification.

Scarred The previous period's consumption enters the persistence term as $\log_{\text{prev}} = \text{logmax}\big(C_{t-1}/\bar{C},\, \alpha_{\text{prev}}\big)$. The income argument to κ_t is the realised previous-period income offset by a calibration wedge, $\mathcal{Y}_t^{\text{signal}} = \max\big(\mathcal{Y}_{t-1} + w,\, \varepsilon_{\mathcal{Y}}\big)$, with $w = \max\big(\mathcal{Y}_0 - \mathcal{Y}_0^{\text{prod}},\, 0\big)$, where $\mathcal{Y}_0^{\text{prod}} = \sum_j(\ell_{j,0} + \Pi_{j,0})$ is the production-side base-year income and $\mathcal{Y}_0$ is the floored calibration baseline defined above, so the wedge is nonzero only where the consumption floor lifts $\mathcal{Y}_0$ above production income.

Frozen Household income is fixed at the base-year level $\mathcal{Y}_0$ for all t, and neither labour income nor profits feed into consumption. The capacity formula becomes $\kappa_t = \max\big((1-s)(1-\nu)\mathcal{Y}_0,\, \varepsilon_\kappa\big)$, constant over time. This mode is useful for holding one region's household demand at its baseline whilst another region is shocked.

In all three cases the updating rule is

$$C_t = \bar{C}\exp\!\left(\rho_1\,\log_{\text{prev}} + \frac{\rho_0}{2}\,\text{logmax}\!\left(\frac{\kappa_t}{\bar{C}},\, \alpha_C^{\text{cur}}\right) + \frac{\rho_0}{2}\,\text{logmax}\!\left(\frac{\kappa_t^{\text{exp}}}{\bar{C}},\, \alpha_C^{\text{exp}}\right)\right), \tag{19}$$

where $\bar{C} = \max(\sum_j c_{j,0}, 1)$ is baseline aggregate consumption, κ_t is current-period capacity, $\kappa_t^{\text{exp}} = (1-s)(1-\nu)\max(\mathcal{Y}_0 \cdot \xi_t, \varepsilon)$ is the expected capacity under expectations ξ_t, and $\alpha_C^{\text{cur}}, \alpha_C^{\text{exp}} \in (0, 1]$ are small floor ratios that prevent the log from becoming unbounded. The result is then clipped to $[\min(\bar{C}\cdot\alpha_C^{\text{exp}},\, \kappa_t),\ \kappa_t]$. Here

$$\rho_0 + \rho_1 = 1. \tag{20}$$

Sectoral consumption demand takes one of two forms. When the subsistence shares vanish ($\sigma_j = 0$, the default), the Cobb-Douglas allocation is used:

$$c_{j,t}^d = \theta_j \cdot C_t \cdot (1 - \epsilon_t), \qquad \theta_j = \frac{c_{j,0}}{\sum_k c_{k,0}}. \tag{21}$$

When subsistence shares > 0 the Stone-Geary Linear Expenditure System (LES) is used instead:

$$c_{j,t}^d = \big[\gamma_j + \beta_j\,(C_t - \Gamma)\big]\,(1 - \epsilon_t), \tag{22}$$

where $\gamma_j = \sigma_j\, c_{j,0}$ is the subsistence floor, $\Gamma = \sum_j \gamma_j$ is total committed expenditure, and $\beta_j = (c_{j,0} - \gamma_j)/(C_0 - \Gamma)$ is the marginal budget share, with $C_0 = \sum_j c_{j,0}$. The subsistence share $\sigma_j \in [0, 1)$ is calibrated as

$$\sigma_j = 1 - \frac{\eta_j}{|\omega|}, \tag{23}$$

where $\eta_j > 0$ is the sector income elasticity and $\omega < 0$ is the Frisch parameter (default $\omega = -3$). Setting $\sigma_j = 0$ everywhere recovers the Cobb-Douglas formula exactly. At base year with $\epsilon = 0$ the LES allocation satisfies $c^d_{j,0} = \gamma_j + \beta_j(C_0 - \Gamma) = c_{j,0}$ identically.

3.3 Inventory Management and Orders

Target inventories are proportional to base-year intermediate flows and a sector-specific coverage n_j (in days):

$$S_{\text{target},ij} = Z_{0,ij} \cdot n_j = A_{ij}\, x_{j,0}\, n_j. \tag{24}$$

Orders for intermediate inputs are

$$O_{ij,t} = A_{ij}\, d_{j,t-1} + \frac{S_{\text{target},ij} - S_{ij,t-1}}{\tau_i}, \quad O_{ij,t} = \max(0, O_{ij,t}), \tag{25}$$

where τ_i is the inventory-adjustment speed of the input row i, so a cell's gap closes at the speed of the input being restocked rather than that of the sector holding it. Each τ_i is sector-specific and lies in an interval $[\underline{\tau}, \overline{\tau}]$ prescribed by the model.

3.4 Production Functions

For the Leontief, adapted Leontief, linear, and CES variants, capacity is determined solely by labour: $x_{\text{cap},j,t} = (\ell_{j,t}/\ell_{j,0})\, x_{j,0}$, and may be further limited by an exogenous output constraint $x^{\text{out}}_{\text{cap},j,t}$, giving an effective bound of $\min(x_{\text{cap},j,t}, x^{\text{out}}_{\text{cap},j,t})$. The KLEMS variant (Section 3.4.5) replaces this scalar bound with a factor bundle that incorporates capital and substitutable intermediates alongside labour. In all cases, production is the minimum of the capacity bound, input-determined output, and demand. The model admits five production types: traditional Leontief, adapted Leontief (with essential inputs), linear, CES, and KLEMS. In every input formula below, the per-cell inventory $S_{ij,t-1}$ is replaced by $S^{\text{eff}}_{ij,t}$ defined in Section 11.1.

3.4.1 Traditional Leontief

$$x_{j,t} = \min\left(\text{capacity}_{j,t},\ \min_{i:A_{ij}>0} \frac{S_{ij,t-1}}{A_{ij}},\ d_{j,t}\right). \tag{26}$$

3.4.2 Adapted Leontief with Essential Inputs

Inputs are split into two groups based on a threshold θ_{ess} (default 0.5):

- **Essential:** $A_{\text{essential},ij} > \theta_{\text{ess}}$ and $A_{ij} > 0$.
- **Non-essential:** $A_{\text{essential},ij} \le \theta_{\text{ess}}$ and $A_{ij} > 0$.

The essential constraint is the usual Leontief minimum over essential inputs only:

$$x^{\text{ess}}_{\text{input},j} = \min_{i:\, A_{\text{essential},ij} > \theta_{\text{ess}},\ A_{ij}>0} \frac{S_{ij,t-1}}{A_{ij}}. \tag{27}$$

The non-essential constraint is an A-weighted average of the per-input capacities over the non-essential set $\mathcal{NE}_j = \{i : A_{\text{essential},ij} \le \theta_{\text{ess}},\ A_{ij} > 0\}$:

$$x^{\text{ne}}_{\text{input},j} = \frac{\sum\limits_{i \in \mathcal{NE}_j} A_{ij}\,\dfrac{S_{ij,t-1}}{A_{ij}}}{\sum\limits_{i \in \mathcal{NE}_j} A_{ij}} = \frac{\sum\limits_{i \in \mathcal{NE}_j} S_{ij,t-1}}{\sum\limits_{i \in \mathcal{NE}_j} A_{ij}}. \tag{28}$$

Each term is $+\infty$ when the corresponding set is empty. The combined input constraint is

$$x_{\text{input},j} = \min\!\big(x^{\text{ess}}_{\text{input},j},\ x^{\text{ne}}_{\text{input},j}\big), \tag{29}$$

and output is

$$x_{j,t} = \min(\text{capacity}_{j,t},\ x_{\text{input},j},\ d_{j,t}). \tag{30}$$

With the default (combined-linkage) method $\mathbf{A}_{\text{essential}}$ is binary $\{0, 1\}$, so all non-zero inputs are either fully essential or fully non-essential.

3.4.3 Linear Production

$$x_{j,t} = \min\left(\text{capacity}_{j,t},\ \frac{\sum_{i=1}^{N} S_{ij,t-1}}{\sum_{i=1}^{N} A_{ij}},\ d_{j,t}\right). \tag{31}$$

3.4.4 CES Production

The CES specification aggregates all inputs i with $A_{ij} > 0$ uniformly, without distinguishing essential from non-essential inputs. Per-input capacities are $q_i = S_{ij,t-1}/A_{ij}$, and the weights are the technical coefficients $w_i = A_{ij}$, normalised to sum to one. With substitution elasticity $\sigma > 0$ and $\rho = (\sigma - 1)/\sigma$:

$$x^{\text{CES}}_{\text{input},j} = \begin{cases} \exp\!\Big(\sum\limits_i \bar{w}_i \log q_i\Big) & \text{if } \sigma = 1, \\ \Big(\sum\limits_i \bar{w}_i\, q_i^{\rho}\Big)^{1/\rho} & \text{otherwise,} \end{cases} \tag{32}$$

where $\bar{w}_i = w_i / \sum_k w_k$ and the sum runs over all i with $A_{ij} > 0$ and finite q_i. Output is then

$$x_{j,t} = \min\!\big(\text{capacity}_{j,t},\ x^{\text{CES}}_{\text{input},j},\ d_{j,t}\big). \tag{33}$$

The default elasticity is $\sigma = 1.5$.

3.4.5 KLEMS Production

The KLEMS variant replaces the scalar labour-capacity bound with a nested factor structure that combines three CES aggregates (for capital-labour, energy inputs, and material inputs) with a Leontief residual for the remaining non-substitutable intermediates.

Sectors are partitioned into three groups by their supply role: $\mathcal{E}$ (energy suppliers), $\mathcal{M}$ (material suppliers), and $\mathcal{D}$ (the Leontief residual, containing all remaining sectors). For sector j define the active input sets $\mathcal{E}_j = \{i \in \mathcal{E} : A_{ij} > 0\}$, $\mathcal{M}_j = \{i \in \mathcal{M} : A_{ij} > 0\}$, $\mathcal{D}_j = \{i \in \mathcal{D} : A_{ij} > 0\}$, and let $q_{ij,t} = S_{ij,t-1}/A_{ij}$ denote the per-input inventory ratio.

With A-normalised weights $\bar{w}^E_{ij} = A_{ij} / \sum_{k \in \mathcal{E}_j} A_{kj}$ and $\rho^E_j = (\sigma^E_j - 1)/\sigma^E_j$, the energy sub-aggregate is

$$E_{j,t} = \begin{cases} \exp\Big(\sum_{i \in \mathcal{E}_j} \bar{w}^E_{ij} \, \log q_{ij,t} \Big) & \sigma^E_j = 1, \\ \Big(\sum_{i \in \mathcal{E}_j} \bar{w}^E_{ij} \, q^{\rho^E_j}_{ij,t} \Big)^{1/\rho^E_j} & \text{otherwise,} \end{cases} \tag{34}$$

with $E_{j,t} = +\infty$ when $\mathcal{E}_j = \emptyset$. The material aggregate $M_{j,t}$ is defined symmetrically with σ^M_j, $\bar{w}^M_{ij}$, and $\mathcal{M}_j$.

For the capital-labour sub-aggregate, labour and capital capacities are

$$x_{\text{cap},L,j,t} = \frac{\ell_{j,t}}{\ell_{j,0}} \, x_{j,0} \, \phi^L_{j,t}, \qquad\qquad x_{\text{cap},K,j,t} = x_{j,0} \, \phi^K_{j,t}, \tag{35}$$

with base-year factor-share weights $w^L_j = \ell_{j,0}/(\ell_{j,0} + \text{cap}_{j,0})$ and $w^K_j = \text{cap}_{j,0}/(\ell_{j,0} + \text{cap}_{j,0})$. The capital-labour aggregate is

$$KL_{j,t} = \begin{cases} x^{w^L_j}_{\text{cap},L,j,t} \, x^{w^K_j}_{\text{cap},K,j,t} & \sigma^{KL}_j = 1, \\ \big(w^L_j \, x^{\rho^{KL}_j}_{\text{cap},L,j,t} + w^K_j \, x^{\rho^{KL}_j}_{\text{cap},K,j,t} \big)^{1/\rho^{KL}_j} & \text{otherwise,} \end{cases} \tag{36}$$

where $\rho^{KL}_j = (\sigma^{KL}_j - 1)/\sigma^{KL}_j$.

The top-level KLE weights are derived from base-year cost shares, $w^{KL}_j \propto (\ell_{j,0} + \text{cap}_{j,0})/x_{j,0}$, $w^E_j \propto \sum_{i \in \mathcal{E}} A_{ij}$, and $w^M_j \propto \sum_{i \in \mathcal{M}} A_{ij}$, normalised to sum to one. With $\rho^{KLE}_j = (\sigma^{KLE}_j - 1)/\sigma^{KLE}_j$,

$$\text{KLE}_{j,t} = \begin{cases} KL^{w^{KL}_j}_{j,t} \; E^{w^E_j}_{j,t} \; M^{w^M_j}_{j,t} & \sigma^{KLE}_j = 1, \\ \big(w^{KL}_j \, KL^{\rho^{KLE}_j}_{j,t} + w^E_j \, E^{\rho^{KLE}_j}_{j,t} + w^M_j \, M^{\rho^{KLE}_j}_{j,t} \big)^{1/\rho^{KLE}_j} & \text{otherwise.} \end{cases} \tag{37}$$

When $w^E_j = 0$ or $w^M_j = 0$, the corresponding term is absent from the aggregate.

The Leontief residual over the remaining inputs is

$$\text{ND}_{j,t} = \min_{i \in \mathcal{D}_j} q_{ij,t}, \tag{38}$$

with $\text{ND}_{j,t} = +\infty$ when $\mathcal{D}_j = \emptyset$.

Output is

$$x_{j,t} = \min\big(\text{KLE}_{j,t}, \; \text{ND}_{j,t}, \; d_{j,t}, \; \bar{x}^{\text{out}}_{j,t} \big). \tag{39}$$

In the labour adjustment of Section 3.1, the potential output is $x^{\text{pot}}_j = \min\big(\text{ND}_{j,t-1}, \, d_{j,t-1}, \, \bar{x}^{\text{out}}_{j,t-1}\big)$: labour adjusts towards the Leontief-constrained potential, whilst energy and material constraints propagate through the KLE channel.

3.5 Rationing

Absent shocks, industries can always meet total demand, i.e. $x_{j,t} = d_{j,t}$. However, in the presence of production capacity and/or input bottlenecks, industries' output may be smaller than total demand (i.e. $x_{j,t} < d_{j,t}$), in which case output is rationed across customers. Simple proportional rationing is assumed. The final delivery from industry i to industry j is the share of orders received,

$$Z_{ij,t} = O_{ij,t}\,\frac{x_{i,t}}{d_{i,t}}. \tag{40}$$

A share of household demand is received,

$$c_{i,t} = c^{d}_{i,t}\,\frac{x_{i,t}}{d_{i,t}}, \tag{41}$$

and the realised final consumption of agents with exogenous final demand is

$$f_{i,t} = f^{d}_{i,t}\,\frac{x_{i,t}}{d_{i,t}}. \tag{42}$$

(When the model disaggregates final demand into government, investment, exports, etc., the same sector-level share $x_{i,t}/d_{i,t}$ applies to each component.) Deliveries are not bounded by inventories. The stock $S_{ij,t-1}$ is held by the purchasing sector j, so a bound on the incoming delivery would preclude the restocking that the order rule requests, and supply feasibility already enters through $x_{i,t}$, which is constrained by the supplier's own input inventories.

3.6 Delivery Allocation

A single allocation step splits feasible output into intermediate deliveries $Z_{ij,t}$, realised household consumption $c_{i,t}$, and one realised array per exogenous final-demand component (government, investment, inventory change, exports, other). Since output never exceeds demand, deliveries sum to feasible output under both rules, $\sum_k Z_{ik,t} + c_{i,t} + \sum_g f_{g,i,t} = x_{i,t}$. Two allocation rules are admitted, distinguished by an intermediate share s^{Z}_{i} and a final share s^{F}_{i}, both clipped to $[0,1]$. Under the proportional rule, $s^{Z}_{i} = s^{F}_{i} = x_{i,t}/d_{i,t}$ per sector i. Under supplier priority, $s^{Z}_{i} = x_{i,t}/\sum_{k=1}^{N} O_{ik,t}$, and the final share is the residual supply/demand ratio

$$s^{F}_{i} = \frac{x_{i,t} - \sum_{k=1}^{N} Z_{ik,t}}{d_{i,t} - \sum_{k=1}^{N} Z_{ik,t}}, \tag{43}$$

output net of intermediate deliveries over demand net of those deliveries. Deliveries are then $Z_{ij,t} = O_{ij,t}\,s^{Z}_{i}$, $c_{i,t} = c^{d}_{i,t}\,s^{F}_{i}$, and $f_{g,i,t} = f^{d}_{g,i,t}\,s^{F}_{i}$ for each component g. Under the direct rationing shock of Section 5 the shocked supplier's row is overridden, $Z_{ij,t} = O_{ij,t}\,\kappa_i$ with capacity fraction κ_i, and household consumption likewise when the shock includes households. The final share is evaluated on the overridden deliveries.

3.7 Inventory Update

The inventory of input i held by sector j is updated according to

$$S_{ij,t} = \max\bigl(0,\; S_{ij,t-1} + Z_{ij,t} - A_{ij}\,x_{j,t}\bigr). \tag{44}$$

In a Leontief production function, where every input is critical, the maximum operator would be redundant because production cannot continue once inventories are run down. The operator is needed for other production functions because industries can produce even after inventories of one or more non-critical inputs are depleted, and the floor at zero prevents inventories from becoming negative.

3.8 Profit and GDP

Profits are computed using labour and value-added shares (capital, tax, import) from the base year:

$$\Pi_{j,t} = x_{j,t} - \sum_{i=1}^{N} Z_{ij,t} - \ell_{j,t} - (\lambda_j^{\text{cap}} + \lambda_j^{\text{tax}} + \lambda_j^{\text{imp}})\, x_{j,t} \;-\; \text{IS}_{j,t} \;+\; \text{REX}_{j,t} \;-\; \text{EPC}_{j,t}. \tag{45}$$

GDP (value added, output minus realised intermediate use, with import adjustment) is

$$\text{GDP}_t = \sum_{j=1}^{N} \Big(x_{j,t} - U_{j,t} - \lambda_j^{\text{imp}}\, x_{j,t} + \text{REX}_{j,t} \Big), \qquad U_{j,t} = \sum_{i=1}^{N} \big(S_{ij,t-1} + Z_{ij,t} - S_{ij,t} \big), \tag{46}$$

where realised use $U_{j,t}$ is taken with $S_{ij,t}$ as defined in the inventory update, before the supplement restore, so $U_{j,t} = \sum_i A_{ij}\, x_{j,t}$ wherever the inventory floor is slack. Value added nets realised use rather than purchases, so inventory accumulation is not booked as value-added loss and the expenditure identity holds with the change in inventories. Supplement purchases enter stock and are expensed on later drawdown, so $\text{IS}_{j,t}$ and $\text{EPC}_{j,t}$ enter the profit identity above but not GDP. $\text{REX}_{j,t}$ is defined in Section 11.

3.9 Savings

Aggregate savings are disposable household income $\mathcal{Y}_t$ (Section 3.2) net of consumption and other household outlays Ω_t,

$$S_t = \mathcal{Y}_t - \sum_{j=1}^{N} c_{j,t} - \Omega_t, \tag{47}$$

where

$$\Omega_t = \frac{\nu}{1-\nu} \sum_{j=1}^{N} c_{j,t} \tag{48}$$

and $\nu \in [0, 1)$ is the extra-expenditure coefficient. When the Keynesian investment closure is active (Section 3.10), the lagged aggregate $\sum_r S_{r,t-1}$ (Section 10.5) feeds directly into investment demand, closing a savings-investment loop.

3.10 Final Demand and Aggregate Demand

Total demand for sector j in period t is

$$d_{j,t} = c_{j,t}^{d} + \sum_{k=1}^{N} O_{jk,t} + f_{\text{gov},j,t} + f_{\text{inv},j,t} + f_{\text{invnt},j,t} + f_{\text{exp},j,t} + f_{\text{other},j,t}, \tag{49}$$

where $c_{j,t}^{d}$ is consumption demand from the Muellbauer block, $\sum_k O_{jk,t}$ is intermediate demand for sector j's output (orders received by sector j from all sectors k), and f_{gov}, f_{inv}, f_{invnt}, f_{exp}, f_{other} are government, investment, inventory (final demand), exports, and any other final demand. The export-pull augmentation $\Delta d_{j,t}^{\text{ep}}$ defined in Section 11.3 is added to $d_{j,t}$ before production solves.

In the standard formulation all non-consumption components are fixed at their base-year vectors: $f_{\text{gov},j,t} = f_{\text{gov},j,0}$ and $f_{\text{inv},j,t} = f_{\text{inv},j,0}$ for all t. The model additionally admits two endogenous rules.

Under the first, government spending is scaled by an aggregate household income index, with any registered exogenous deviation added on top:

$$f_{\text{gov},j,t} = f_{\text{gov},j,0}\left[1 + \eta^{\text{gov}}\left(\frac{\mathcal{Y}_{t-1}}{\mathcal{Y}_0} - 1\right)\right] + \left(\tilde{f}_{\text{gov},j,t} - f_{\text{gov},j,0}\right), \tag{50}$$

where $\eta^{\text{gov}} \in \mathbb{R}$ is the income elasticity of government spending, $\mathcal{Y}_{t-1} = \sum_r \mathcal{Y}_{r,t-1}$ is lagged aggregate household income (Section 10.3), $\mathcal{Y}_0 = \sum_r \mathcal{Y}_{0,r}$ is its base-year value, and $\tilde{f}_{\text{gov},j,t}$ is the registered exogenous government path, equal to $f_{\text{gov},j,0}$ when no shock is registered. Setting $\eta^{\text{gov}} = 0$ recovers the exogenous-government case exactly.

Under the second, a Keynesian closure, investment demand scales with smoothed lagged aggregate household savings. With smoothing weight $w \in (0, 1]$, adjustment speed $\alpha \in (0, 1]$ and growth cap κ,

$$\bar{S}_t = w\, S_{t-1} + (1 - w)\, \bar{S}_{t-1}, \qquad \sigma_t = \max\left(\frac{\bar{S}_t}{S_0},\ 0\right), \tag{51}$$

$$\lambda_t = \begin{cases} \min\big(\hat{\lambda}_t,\ \lambda_{t-1}(1+\kappa)\big), & \lambda_{t-1} > 0, \\ \hat{\lambda}_t, & \lambda_{t-1} = 0, \end{cases} \qquad \hat{\lambda}_t = \max\big(\lambda_{t-1} + \alpha\,(\sigma_t - \lambda_{t-1}),\ 0\big), \tag{52}$$

$$f_{\text{inv},j,t} = f_{\text{inv},j,0}\,\lambda_t + \left(\tilde{f}_{\text{inv},j,t} - f_{\text{inv},j,0}\right), \tag{53}$$

where $S_{t-1} = \sum_r S_{r,t-1}$ is lagged aggregate household savings (Section 3.9 / 10.5), S_0 is its base-year value, $\bar{S}_0 = S_0$, $\lambda_0 = 1$, and $\tilde{f}_{\text{inv},j,t}$ is the registered exogenous investment path, equal to $f_{\text{inv},j,0}$ when no shock is registered. The cap is not applied from $\lambda_{t-1} = 0$, which it would otherwise hold at zero for the remainder of the horizon. When κ is unset the first branch reduces to the second. Negative aggregate savings is absorbed by the floor on σ_t, so with $\alpha \in (0, 1]$ and $\lambda_0 = 1$ the floor on $\hat{\lambda}_t$ is a guard that never binds. The cap binds the endogenous scale only, never the exogenous deviation. Both rules use only lagged $(t-1)$ quantities and therefore preserve the recursive causal order of Section 8.

4 Exogenous Consumption Shock

An exogenous reduction in household expenditure is modelled by setting the shock intensity $\epsilon_t \in [0, 1)$ for any subset of periods. The factor $(1 - \epsilon_t)$ multiplies the full sectoral demand expression from Section 3.2, whether the Cobb-Douglas or LES allocation is active. When demand is the binding constraint, all production functions yield identical results. Divergence arises only when supply constraints bind.

5 Exogenous Supply Shocks

Three sector-level supply-side mechanisms are admitted.

Under an output-constraint shock, the output of sector i^* is capped at a fraction of its base-year level for each period t in the shock window:

$$\bar{x}^{\text{out}}_{i^*,t} = (1 - \zeta_{i^*,t})\, x_{i^*,0}, \qquad \zeta_{i^*,t} \in [0, 1), \tag{54}$$

so the effective capacity in Section 3.4 is $\min(x_{\text{cap},j,t},\, \bar{x}^{\text{out}}_{j,t})$ for all j.

Under a rationing shock, the output of sector i^* is capped at a fraction $\eta_{i^*,t} \in (0, 1]$ of base-year output, $\bar{x}^{\text{out}}_{i^*,t} = \eta_{i^*,t}\, x_{i^*,0}$, and all deliveries from i^* are simultaneously rationed by the same factor, overriding the allocation rule of Section 3.6:

$$Z_{i^*j,t} = \eta_{i^*,t}\, O_{i^*j,t}. \tag{55}$$

Final demand from i^* to households is optionally rationed by the same factor.

Under a factor-productivity shock, the labour and capital productivities of sector j^* are scaled by $\phi^L_{j^*,t}$ and $\phi^K_{j^*,t}$ respectively at and after period t^*. Effective labour capacity becomes

$$x_{\text{cap},j^*,t} = \frac{\ell_{j^*,t}}{\ell_{j^*,0}}\, x_{j^*,0}\, \phi^L_{j^*,t}, \tag{56}$$

and the capital cost share in the profit identity (Section 3.8) becomes $\lambda^{\text{cap}}_{j^*}/\phi^K_{j^*,t}$. Values $\phi > 1$ represent factor-augmenting technical change, whilst $\phi < 1$ represents efficiency loss.

Downstream shortages propagate through reduced deliveries and inventory drawdown at receiving sectors. The production function determines the severity. Under traditional Leontief every input is binding. Under linear production aggregate substitution dampens the effect. Under adapted Leontief propagation depends on whether the shocked input is classified as essential.

6 Parameter Uncertainty

The structural parameters $\{\rho_1,\, \gamma_{\text{hire},j},\, \gamma_{\text{fire},j},\, \tau_j,\, s_r,\, \beta_r\}$ are uncertain. Sector-level parameters (ρ_1, $\gamma_{\text{hire},j}$, $\gamma_{\text{fire},j}$, τ_j) are drawn once per simulation, whilst per-region parameters (s_r, β_r) are drawn independently for each region. Uncertainty is propagated by drawing D independent parameter vectors from prescribed distributions, solving the model for each draw, and characterising the resulting distribution of output quantities. The trade-closure parameters $\{\text{flex}_i,\, \bar{q}_{j'},\, \text{epull}_j\}$ are treated as exogenous calibration and are not perturbed. The price pass-through fractions ψ^+, ψ^- (Section 12) are likewise exogenous and not perturbed. Because the same parameter vector governs all periods, uncertainty bands may persist or widen beyond the shock window. They reflect structural parameter uncertainty, not uncertainty about the shock itself.

7 Model Assumptions and Accounting Consistency

The row identity of the base-year accounts ($\mathbf{x}_0 = \mathbf{Z}_0\mathbf{1} + \mathbf{f}_0$) and the column identity (output equals domestic intermediate inputs plus imports plus value added) are assumed to hold. Simulated flows satisfy these identities by construction under both allocation rules, since output never exceeds demand. Only in periods where the direct rationing shock overrides deliveries can the realised split deviate from feasible output. The model does not impose market-clearing in the Walrasian sense: output is quantity-determined as the minimum of capacity, input availability and demand, so excess demand can persist. When the price pass-through of Section 12 is active, the price index $\mathbf{p}_t$ is a cost-push overlay: it does not enter the row or column identities and does not alter quantity determination, with the single exception of the optional household real-income deflation defined there.

8 Causal Structure

The model is solved sequentially within each period:

1. Labour $\ell_{j,t}$ is adjusted from the previous period's realised constraints, giving current-period capacity $x_{\mathrm{cap},j,t}$.
2. Household consumption C_t (Muellbauer rule) and intermediate orders $O_{ij,t}$ (inventory-replenishment rule) are determined independently of production. When the deflation of Section 12 is active, the income signal entering C_t is divided by the lagged regional price index $P_{r,t-1}$.
3. Aggregate demand $d_{j,t}$ is assembled (Section 3.10) and augmented by the export-pull supplement $\Delta d^{\mathrm{ep}}_{j,t}$ (Section 11.3). Where the endogenous rules of Section 3.10 are active, $f_{\mathrm{gov},j,t}$ and $f_{\mathrm{inv},j,t}$ are evaluated from $\mathcal{Y}_{r,t-1}$ and $S_{r,t-1}$ respectively, both computed in the latter stages of the preceding period.
4. Production $x_{j,t}$ is the minimum of capacity, input availability, and (augmented) demand, with the input cell $S_{ij,t-1}$ replaced by $S^{\mathrm{eff}}_{ij,t}$ when import flexibility is active (Section 11.1).
5. The supplement actually used yields $\mathrm{IS}_{j,t}$ and the sourced part is credited to RoW as $\mathrm{REX}_{j',t}$ (Section 11.2).
6. Feasible output is split into intermediate deliveries $Z_{ij,t}$, realised consumption $c_{i,t}$, and the realised exogenous final-demand components by the allocation rule of Section 3.6.
7. Inventories are updated, with the consumed supplement added back to $S_{ij,t}$.
8. Profits, regional GDP, and savings are computed. The export-pull correction $\mathrm{EPC}_{j,t}$ is applied to profits (Section 11.4).
9. When price pass-through is active, the price index $\mathbf{p}_t$ is updated from the unit-cost shocks in force at t (Section 12); through the one-period lag it enters the income signal of the following period.

The ordering is recursive rather than simultaneous, and disequilibrium between supply and demand is the normal state.

9 Time-Varying Technical Coefficients

The standard formulation assumes a fixed technical coefficient matrix $\mathbf{A}$. The model also supports a piecewise-constant trajectory $\{\mathbf{A}_t\}$ in which $\mathbf{A}$ is replaced by a new matrix $\mathbf{A}'$ at one or more specified periods. The trajectory is defined by a finite set of change periods $\mathcal{T}_{\mathrm{ch}} \subset \{1, \ldots, T-1\}$, and for each $t^* \in \mathcal{T}_{\mathrm{ch}}$ a matrix $\mathbf{A}^{(t^*)}$ is prescribed. The active coefficient matrix at period t is

$$\mathbf{A}_t = \mathbf{A}^{(t^*_{\mathrm{last}})}, \qquad t^*_{\mathrm{last}} = \max\{t^* \in \mathcal{T}_{\mathrm{ch}} : t^* \le t\}, \tag{57}$$

with $\mathbf{A}_t = \mathbf{A}$ (the base-year matrix) when $\mathcal{T}_{\mathrm{ch}}$ is empty or $t < \min \mathcal{T}_{\mathrm{ch}}$.

At each change period t^*, the inventory target is updated to reflect the new technology:

$$S^{(t^*)}_{\mathrm{target},ij} = A^{(t^*)}_{ij}\, x_{j,0}\, n_j, \tag{58}$$

and the essential-input classification $\mathbf{A}_{\mathrm{essential}}$ is recomputed via the combined-linkage method applied to $\mathbf{A}^{(t^*)}$ with base-year gross output $\mathbf{x}_0$ (adapted Leontief only). Under the KLEMS variant,

the sector partition $\mathcal{E}$, $\mathcal{M}$, $\mathcal{D}$ is invariant to changes in $\mathbf{A}$, and the KLE weights remain calibrated to the base-year matrix $\mathbf{A}$. When price pass-through is active, the two cost-push inverses are also recomputed from the new matrix, $\mathbf{L}_t^{\pm} = (\mathbf{I} - \psi^{\pm}\mathbf{A}^{(t^*)})^{-1}$ (Section 12), so that price propagation reflects the current structure.

10 Multi-Region Formulation

10.1 Setup and Notation

The model spans R regions, each containing N_r sectors, for a total of $N = \sum_{r=1}^{R} N_r$ sectors indexed globally. Region r contains the sector index set $J_r \subset \{0, \ldots, N-1\}$, where the sets J_r partition $\{0, \ldots, N-1\}$ without overlap. The region membership of each sector is encoded in the mapping $r : \{0, \ldots, N-1\} \to \{0, \ldots, R-1\}$, supplied with the data.

Per-region household accounting, per-region consumption aggregates, and a regional GDP decomposition are added. All sector-level equations from Sections 1 to 9 apply without modification to the full N-sector system spanning all regions. The trade closures of Section 11 require a designated RoW region, and are inert in single-region runs.

10.2 Data Structure

All final demand categories are (N, R) matrices. The (j, r) entry records the base-year spending of region r's agents on sector j's output, and sector j may belong to any region. For region r the five vectors are

- $\mathbf{c}_r^{(0)} \in \mathbb{R}^N$, household consumption,
- $\mathbf{g}_r \in \mathbb{R}^N$, government demand,
- $\mathbf{v}_r \in \mathbb{R}^N$, gross fixed capital formation,
- $\boldsymbol{\iota}_r \in \mathbb{R}^N$, changes in inventories and valuables,
- $\mathbf{e}_r \in \mathbb{R}^N$, exports to rest-of-world.

The row identity for sector j is

$$x_{j,0} = \sum_{k=1}^{N} Z_{0,jk} + \sum_{r=1}^{R} \big(c_{j,r}^{(0)} + g_{j,r} + v_{j,r} + \iota_{j,r} + e_{j,r}\big). \tag{59}$$

The technical coefficient matrix is constructed from the full $N \times N$ flow matrix $\mathbf{Z}_0$, exactly as in the single-region case: $A_{ij} = Z_{0,ij}/x_{j,0}$. The column-viability condition $\sum_{i=1}^{N} A_{ij} < 1$ is enforced across all supplying regions.

Inter-regional intermediate trade is encoded in the off-diagonal blocks of $\mathbf{Z}_0$: for $i \in J_r$, $j \in J_s$ with $r \neq s$, the entry $Z_{0,ij}$ is the base-year intermediate supply from sector i in region r to sector j in region s. These flows are handled automatically by the full $N \times N$ matrix operations and must not be included in the rest-of-world import vector, which is reserved for imports from outside the modelled region system.

10.3 Per-Region Household Accounting

Each region r has independent household-accounting scalars derived from base-year data:

$$\bar{C}_r = \sum_{j=1}^{N} c_{j,r}^{(0)}, \qquad \theta_{j,r} = \frac{c_{j,r}^{(0)}}{\bar{C}_r}, \tag{60}$$

$$\mathcal{Y}_{0,r}^{\text{prod}} = \sum_{j \in J_r} \ell_{j,0} + \sum_{j \in J_r} \Pi_{j,0}, \qquad \mathcal{Y}_{0,r} = \max\Big(\mathcal{Y}_{0,r}^{\text{prod}},\ \frac{\bar{C}_r}{1-\nu_r}\Big), \tag{61}$$

$$s_r = 1 - \frac{\bar{C}_r}{(1-\nu_r)\,\mathcal{Y}_{0,r}}, \tag{62}$$

where $\nu_r \in [0,1)$ is the region-r extra-expenditure coefficient, $\mathcal{Y}_{0,r}^{\text{prod}}$ is production-side base income, $\mathcal{Y}_{0,r}$ is the calibration baseline floored at the income needed to fund base-year consumption, and s_r is the inferred base-year savings rate, zero where the floor binds. When skill-stratified savings rates are active, s_r is replaced by the skill-income-weighted effective rate

$$s_r^{\text{eff}} = 1 - \sum_{s \in \{L,M,H\}} (1 - s_s)\,\varphi_{s,r}, \tag{63}$$

where s_s is the savings rate for skill tier $s \in \{\text{Low, Medium, High}\}$ and $\varphi_{s,r} = \sum_{j \in J_r} \ell_{s,j,0} \,/\, \sum_{j \in J_r} \ell_{j,0}$ is the base-year skill income share for region r, with $\ell_{s,j,0}$ the base-year wage bill attributed to tier s in sector j. By default this feature is inactive and s_r from the base-year calibration applies.

Household income for region r at period t is

$$\mathcal{Y}_{r,t} = \max\Big(\beta_r \sum_{j \in J_r} \ell_{j,0} + (1-\beta_r) \sum_{j \in J_r} \ell_{j,t} + \sum_{j \in J_r} \Pi_{j,t},\ \varepsilon_{\mathcal{Y}}\Big), \tag{64}$$

where $\beta_r \in [0,1]$ is the region-r benefits parameter, applied to labour income only whilst profit enters in full. The income signal passed to the consumption function depends on the household closure mode selected for region r. Under *return-to-base* it is $\mathcal{Y}_{0,r}\,\xi_t$. Under *scarred* it is $\mathcal{Y}_{r,t-1} + w_r$, where the calibration wedge $w_r = \max\big(\mathcal{Y}_{0,r} - \mathcal{Y}_{0,r}^{\text{prod}},\, 0\big)$ closes the structural identity gap (see Section 3.2). Under *frozen* it is $\mathcal{Y}_{0,r}$, independent of t. When the optional price deflation of Section 12 is active, this signal is divided by the lagged regional price index $P_{r,t-1}$ to express it in real terms.

10.4 Per-Region Consumption and Aggregate Demand

The aggregate consumption for region r at period t, $C_{r,t}$, is determined by the same log-linear Muellbauer rule as in the single-region case (Section 3.2), with all scalars replaced by their region-r counterparts: $\bar{C}_r$, $\kappa_{r,t}$, s_r, ν_r, and the region-specific consumption shock $\epsilon_{r,t} \in [0,1)$.

Sectoral desired consumption demand is formed by summing over regions. In the Cobb-Douglas case:

$$c_{j,t}^{d} = \sum_{r=1}^{R} \theta_{j,r}\, C_{r,t}\,(1 - \epsilon_{r,t}), \qquad \theta_{j,r} = \frac{c_{j,r}^{(0)}}{\bar{C}_r}. \tag{65}$$

When LES is active, the per-region subsistence floor and marginal budget share are computed from the region-r base consumption vector: $\gamma_{j,r} = \sigma_j\, c_{j,r}^{(0)}$, $\Gamma_r = \sum_j \gamma_{j,r}$, and $\beta_{j,r} = (c_{j,r}^{(0)} - \gamma_{j,r})/(\bar{C}_r -$

Γ_r). The per-region contribution is then $[\gamma_{j,r} + \beta_{j,r}(C_{r,t} - \Gamma_r)](1 - \epsilon_{r,t})$, aggregated over r as before. In both cases $\theta_{j,r}$ (or $\gamma_{j,r}$, $\beta_{j,r}$) is defined over all N sectors, not only those in J_r, reflecting the fact that region r's households may purchase from any sector in the economy.

Final demand components (government, investment, inventories, exports) are aggregated across regions into a single N-vector for each period and enter the aggregate demand equation exactly as in the single-region model; the government and investment components are endogenous when the rules of Section 3.10 are in use.

10.5 Per-Region Savings

Aggregate savings for region r at period t are

$$S_{r,t} = \mathcal{Y}_{r,t} - \sum_{j=1}^{N} c_{j,r,t} - \frac{\nu_r}{1 - \nu_r} \sum_{j=1}^{N} c_{j,r,t}, \tag{66}$$

where $c_{j,r,t} = \theta_{j,r}\, C_{r,t} \cdot (x_{j,t}/d_{j,t})$ is realised consumption of sector j attributed to region r. The economy-wide aggregate $S_t = \sum_r S_{r,t}$ is consistent with the single-region formula.

10.6 Regional GDP Decomposition

Regional GDP is value added net of rest-of-world imports and the sourced-supplement credit, summed over sectors belonging to region r,

$$\mathrm{GDP}_{r,t} = \sum_{j \in J_r} \left(x_{j,t} - U_{j,t} - \lambda_j^{\mathrm{imp}}\, x_{j,t} + \mathrm{REX}_{j,t} \right). \tag{67}$$

The realised-use deduction $U_{j,t}$ (Section 3.8) spans all supplying sectors across all regions, so cross-region intermediate flows are correctly netted out. The import share $\lambda_j^{\mathrm{imp}} = m_{j,0}/x_{j,0}$, where $m_{j,0}$ is sector j's base-year flow from outside the modelled region system, covers only those external imports. REX is defined in Section 11. By construction, $\sum_r \mathrm{GDP}_{r,t} = \mathrm{GDP}_t$.

10.7 Inter-Regional Trade Balance

The intermediate export flow from region r to region s in period t is

$$M_{rs,t} = \sum_{i \in J_r} \sum_{j \in J_s} Z_{ij,t}, \quad r \neq s. \tag{68}$$

The trade balance of region r is

$$\mathrm{TB}_{r,t} = \sum_{s \neq r} M_{rs,t} - \sum_{s \neq r} M_{sr,t}. \tag{69}$$

By accounting, $\sum_r \mathrm{TB}_{r,t} = 0$ for every t. $M_{rs,t}$ captures intermediate orders only. The trade-closure flows (sourced supplement $\mathrm{REX}_{j',t}$ and export-pull augmentation $\Delta d_{j,t}^{\mathrm{ep}}$) are recorded separately and are not summed into M.

10.8 Per-Region Monte Carlo Sampling

In the Monte Carlo analysis (Section 6), the benefits parameter β_r and savings rate s_r are drawn independently per region from their respective distributions. The consumption-persistence and labour-adjustment parameters remain sector-level draws, unchanged from the single-region case. For each simulation draw, the derived household accounting scalars $\bar{C}_r$, $\kappa_{r,0}$, and $\mathcal{Y}_{0,r}$ are refreshed to be consistent with the sampled β_r and s_r before the model is solved. Regional value added $\mathrm{GDP}_{r,t}$ is recorded for all r and t across simulation draws, alongside the economy-wide aggregate GDP_t.

10.9 Single-Region Special Case

When $R = 1$, the multi-region formulation reduces exactly to the single-region model. All per-region quantities collapse to scalars: $\theta_{j,0} = \theta_j$, $\epsilon_{0,t} = \epsilon_t$, and $\mathrm{GDP}_{0,t} = \mathrm{GDP}_t$. The accounting identities and order of operations are unchanged. The optional price pass-through (Section 12) applies unchanged, with the regional price index $P_{r,t}$ reducing to a single economy-wide index.

11 Trade Closures

The trade closures augment the static rest-of-world (RoW) import leakage λ_j^{imp} with three quantity-side mechanisms, namely an inventory supplement, a RoW-sourcing credit, and a CET-analogue export pull. The model has no price-based trade mechanism, so Armington and CET in their CGE forms are not represented. The closures below are the quantity-side analogues.

Two auxiliary objects are defined at model initialisation. Let r_{RoW} denote the index of the rest-of-world region and J_{RoW} its sector set. Each sector j is labelled $(r(j), n(j))$, the pair of its region and sector name. The same-good RoW index is

$$g(j) = \begin{cases} j' \in J_{\mathrm{RoW}} \text{ such that } n(j') = n(j) & \text{if such } j' \text{ exists,} \\ -1 & \text{otherwise.} \end{cases}$$

The named-region sector mask is $J_{\mathrm{named}} = \{0, \ldots, N-1\} \setminus J_{\mathrm{RoW}}$.

11.1 Import Flexibility (Inventory Supplement)

For each input i and using sector j, the inventory available to production is augmented by a fraction $\mathrm{flex}_i \in [0, 1]$ of any shortfall against the target $S_{ij}^{\mathrm{target}} = A_{ij}\, x_{j,0}\, n_j$,

$$\Delta S_{ij,t} = \mathrm{flex}_i \cdot \max\bigl(S_{ij}^{\mathrm{target}} - S_{ij,t-1},\, 0\bigr), \qquad S_{ij,t}^{\mathrm{eff}} = S_{ij,t-1} + \Delta S_{ij,t}. \tag{70}$$

Let $x_{j,t}^{\mathrm{ns}}$ denote the output sector j would produce if the input constraint in Section 3.4 were evaluated with $S_{ij,t-1}$ instead of $S_{ij,t}^{\mathrm{eff}}$ (all other inputs equal). The supplement is charged only on sectors where it raised output:

$$\Delta S_{ij,t}^{\mathrm{benefit}} = \Delta S_{ij,t} \cdot \mathbb{1}\bigl[x_{j,t} > x_{j,t}^{\mathrm{ns}}\bigr], \qquad \mathrm{IS}_{j,t} = \sum_{i=1}^{N} \Delta S_{ij,t}^{\mathrm{benefit}}. \tag{71}$$

After production, the consumed supplement is added back to $S_{ij,t}$ so that domestic inventories are not eroded by the supplement.

11.2 Sourced Supplement (Crediting RoW)

The supplement of Section 11.1 is attributed to RoW's same-good sector via $g(\cdot)$ and credited to its profit and GDP. The raw revenue claim per RoW sector $j' \in J_{\text{RoW}}$ is

$$Q_{j',t} = \sum_{i:\, g(i)=j'} \sum_{k=1}^{N} \Delta S^{\text{benefit}}_{ik,t}, \tag{72}$$

capped at a fraction $\bar{q}_{j'} \in [0,1]$ of that RoW sector's current output:

$$\text{REX}_{j',t} = \min\big(Q_{j',t},\ \bar{q}_{j'}\, x_{j',t}\big), \qquad j' \in J_{\text{RoW}}, \tag{73}$$

with $\text{REX}_{j,t} = 0$ for $j \notin J_{\text{RoW}}$. The uncapped remainder $Q_{j',t} - \text{REX}_{j',t}$ together with the supplement on inputs i with $g(i) = -1$ remains an unsourced injection.

11.3 Export Pull (CET-Analogue)

A fraction $\text{epull}_j \in [0,1]$ of last period's slack capacity in a named-region sector is absorbed by external demand. The capacity proxy is

$$\tilde{x}_{\text{cap},j,t-1} = \begin{cases} x_{\text{cap},j,t-1} & \text{if } \ell_{j,0} > 0, \\ x_{j,0} & \text{otherwise,} \end{cases} \tag{74}$$

and the per-period slack, clipped at $x_{j,0}$, is

$$\eta_{j,t} = \min\Big(\max\big(\tilde{x}_{\text{cap},j,t-1} - x_{j,t-1},\, 0\big),\ x_{j,0}\Big). \tag{75}$$

The additional demand on $d_{j,t}$ is

$$\Delta d^{\text{ep}}_{j,t} = \text{epull}_j \cdot \eta_{j,t} \cdot \mathbb{1}[j \in J_{\text{named}}]. \tag{76}$$

11.4 Export-Pull Value-Added Correction

The export-pull supplement adds mass to $x_{j,t}$ without flowing through orders to suppliers, so $\sum_i Z_{ij,t}$ does not rise to match the extra production. GDP needs no correction, realised use carrying the input draw of the extra output. The deduction restoring the profit identity is

$$\text{EPC}_{j,t} = \Big(\sum_{i=1}^{N} A_{ij}\Big)\, \Delta d^{\text{ep}}_{j,t}, \tag{77}$$

with $\text{EPC}_{j,t} = 0$ for $j \notin J_{\text{named}}$.

12 Endogenous Price Pass-Through

This section defines an optional price channel layered on the quantity model. It is inactive by default, in which case $\mathbf{p}_t = \mathbf{1}$ for all t and nothing below alters the formulation of Sections 1 to 11. When active it produces a cost-push price index, and, as a separate option, couples that index back to household demand.

12.1 Cost-Push Price Index

Let $\psi^+ \in [0,1]$ and $\psi^- \in [0,1]$ be the pass-through fractions for upward and downward unit-cost changes. Two cost-push Leontief inverses are formed from the technical coefficient matrix,

$$\mathbf{L}^+ = (\mathbf{I} - \psi^+ \mathbf{A})^{-1}, \qquad \mathbf{L}^- = (\mathbf{I} - \psi^- \mathbf{A})^{-1}, \tag{78}$$

which reduce to the Leontief inverse $\mathbf{L}$ of Section 1.2 when the corresponding pass-through fraction equals one. The transpose governs forward cost transmission, so that a unit-cost change in a sector raises the price of the sectors that use its output, in the standard cost-push form [5].
A direct unit-cost shock k on sector i_k is a fractional change $\chi_k > -1$ of base unit cost, active over the window $t_k^{\mathrm{s}} \le t < t_k^{\mathrm{e}}$, with $t_k^{\mathrm{e}} = T$ giving a permanent level shift. The active shock vector $\boldsymbol{\chi}_t$ aggregates, for each sector, all shocks in force,

$$\chi_{i,t} = \sum_{k\,:\,i_k = i} \chi_k \mathbb{1}[t_k^{\mathrm{s}} \le t < t_k^{\mathrm{e}}], \tag{79}$$

and is split by sign, $\chi_{i,t}^+ = \max(\chi_{i,t}, 0)$ and $\chi_{i,t}^- = \min(\chi_{i,t}, 0)$. The price index is the cost-push level implied by the active shocks, floored to remain positive,

$$p_{j,t} = \max\Big(1 + \big(\mathbf{L}^{+\top} \boldsymbol{\chi}_t^+ + \mathbf{L}^{-\top} \boldsymbol{\chi}_t^-\big)_j,\ p_{\min}\Big), \qquad p_{j,0} = 1, \tag{80}$$

with $p_{\min} > 0$ a small floor. Because the level is rebuilt each period from the shocks then in force, a finite window returns the index to its base once it closes, whilst a permanent shock holds the shifted level. The asymmetry is attributed by the sign of the originating shock rather than by the net direction of each sector's price, which is the exact consequence of the linear split into $\boldsymbol{\chi}_t^+$ and $\boldsymbol{\chi}_t^-$.
Under piecewise-constant technical change (Section 9) the inverses $\mathbf{L}^{\pm}$ are recomputed from the active matrix, so that propagation reflects the current structure.

12.2 Household Real-Income Deflation

As a separate option, the price index deflates household demand. The regional consumption price index is the consumption-share-weighted price,

$$P_{r,t} = \sum_{j=1}^{N} \theta_{j,r}\, p_{j,t}, \qquad P_{r,0} = 1, \tag{81}$$

where $\theta_{j,r}$ are the base-year consumption shares of Section 10, which satisfy $\sum_j \theta_{j,r} = 1$, so $P_{r,0} = 1$. The income signal entering the consumption function for region r is divided by the previous period's index,

$$\tilde{\mathcal{Y}}_{r,t}^{\mathrm{signal}} = \frac{\mathcal{Y}_{r,t}^{\mathrm{signal}}}{\max(P_{r,t-1},\, \varepsilon_P)}, \tag{82}$$

with $\varepsilon_P > 0$ a numerical floor. The one-period lag uses prices already determined at $t-1$, and so avoids the within-period circularity that a contemporaneous index would introduce, given the recursion of Section 8. This deflation is the only channel through which the price index feeds back into the quantity model. When it is inactive the income signal is used in nominal terms exactly as in Sections 3.2 and 10.

13 References

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